U-VOWEL: Uncertainty-Routed VLA and Video-World-Model Policies for Out-of-Distribution Robot Manipulation

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Abstract— **Generalist robot policies can follow diverse language instructions yet their reliability falls under changes in objects, scenes and task descriptions. This paper studies whether two action-generating policy families can compensate for different failures: a fast vision-language-action policy and a video-world-model policy that decodes actions from a video-trained predictive representation. We present U-VOWEL, a runtime framework that instruments each branch with native uncertainty evidence, maps recent evidence to a branch-specific failure score and routes control under a planning-cost budget. The controller uses the VLA as its fast path, invokes the video-world-model policy selectively and discards rejected proposals before any later fresh-state query. In the current single-seed evaluation on four selected LIBERO-PRO Goal perturbation suites, the selected controller completes 888 of 2,000 episodes, compared with 836 for the standalone VLA and 812 for the standalone video-world-model policy. Its mean wall time over successful episodes is 7.35 seconds, between the VLA's 2.69 seconds and the video-world-model policy's 14.16 seconds. The gains are concentrated in language and object perturbations, while swap and task perturbations remain difficult. These results motivate uncertainty-guided policy routing as a compute-aware approach to out-of-distribution manipulation and identify the failure regimes that still require new policy capability.**

I. Introduction

Robot manipulation integrates perception, planning and control to modify the physical environment [1]. A deployed robot must preserve these capabilities as objects, workspaces and language instructions change. Learning from demonstrations provides a practical route to such behavior, yet training can cover only a limited subset of the states encountered during deployment. Small action errors may move the robot outside this subset and compound over a closed-loop rollout [2]. Generalization is therefore sequential: the robot must interpret an unfamiliar situation and keep its trajectory recoverable as every action changes the next observation.

Vision-language-action (VLA) models expand the range of situations that one policy can interpret. VLA architectures have progressed from language-conditioned transformers toward VLM-based policies and continuous action generation. Representative developments include RT-1, RT-2, OpenVLA, π_0 and OpenVLA-OFT [3]. Our fast branch follows SimVLA, a compact architecture that decouples perception from action generation [4]. These models condition direct action prediction on visual and language inputs. Their breadth does not establish robustness under every evaluation shift. LIBERO-PRO reports substantial degradation under controlled perturbations to object appearance, object position, semantic instructions, task definitions and environments [5].

This gap exposes a missing deployment capability. Producing an action does not tell the controller whether that action is supported by familiar evidence or whether the policy is entering a failure regime. A generative action distribution may expose likelihood, variance or sample disagreement, but these quantities are not automatically calibrated probabilities of task failure. In VLAs, confidence calibration varies across action dimensions and over the task horizon [6], while per-step and per-degree-of-freedom entropy can contain brief failure-related spikes [7]. A robot therefore needs an operational way to recognize when its current proposal may be unreliable. Without this signal, it cannot decide when continued execution is justified, when a shorter replanning horizon is appropriate or when another policy should be consulted.

An uncertainty estimate becomes useful when it changes the control decision. A rejected proposal supplies no alternative behavior, whereas a second policy can propose actions through a different predictive process. Recent video-based robot policies show that predicted visual futures can support executable behavior [8], [9], [10]. This development motivates pairing a fast direct policy with a more expensive predictive policy.

This pairing creates a selective-control problem. A fixed policy choice cannot respond when reliability changes during an episode, whereas querying both policies at every decision incurs the cost of predictive generation even when the direct policy is reliable. The two branches also expose uncertainty evidence with different meanings and scales, which prevents a direct comparison of their raw signals.

We present U-VOWEL, an uncertainty-guided controller that uses the VLA as its fast default policy and selectively queries the video-world-model policy when the VLA proposal appears unreliable. Branch-specific temporal monitors convert each policy's uncertainty evidence into routing decisions while limiting use of the more expensive predictive branch (Fig. 1).

The paper makes four contributions:

(i) We introduce a heterogeneous closed-loop controller that routes between independently action-generating vision-language-action and video-world-model policies.

(ii) We instrument both policies with uncertainty evidence. NLL-derived objectives train a heteroscedastic velocity-variance head for the VLA and a residual-variance head for video prediction, complemented by candidate disagreement and branch-native action or prediction statistics.

(iii) We develop branch-specific temporal risk monitors that combine recent uncertainty evidence into failure scores and apply separately calibrated operating thresholds to the two policies.

(iv) We introduce a selective-compute execution strategy

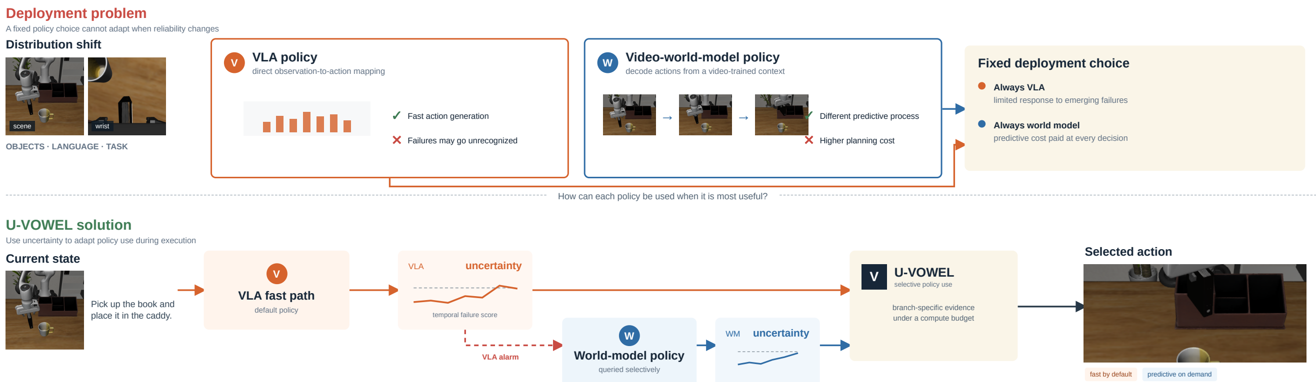

Fig. 1. Motivation for U-VOWEL. Under distribution shift, a direct VLA provides fast control but may not recognize emerging failures, while an always-on video-world-model policy incurs predictive planning cost at every decision. U-VOWEL uses branch-specific uncertainty to retain the VLA fast path and invoke the predictive policy selectively.

that keeps the VLA as the fast path, invokes the video-world-model policy after a VLA alarm and bounds repeated predictive queries through fresh-state escalation and a finite latch.

II. Related Work

A. Manipulation learning and generalist policies

Imitation learning addresses sequential distribution mismatch primarily during training. DAgger collects expert labels on states induced by the learner and adds them to the training set [2]. This exposes the policy to recovery states that are absent from the original demonstrations, but requires interactive expert supervision. It does not provide a confidence signal for a frozen policy at deployment.

Action-sequence policies address a different aspect of closed-loop behavior: the temporal granularity of prediction and execution. ACT uses a conditional variational autoencoder to generate action chunks and temporally ensembles overlapping predictions [11]. Diffusion Policy represents multimodal action sequences through conditional denoising and executes them in a receding-horizon loop [12]. Chunking moves the unit of prediction from one action to a sequence and can smooth local behavior, while making the whole proposed sequence the relevant unit of reliability.

Generalist VLA policies place semantic task interpretation and continuous control within one learned pipeline [3]. Their episode outcomes consequently combine several capabilities, including visual grounding, instruction interpretation and action generation. LIBERO evaluates transfer across lifelong manipulation tasks, while LIBERO-PRO applies controlled distribution shifts to the benchmark [13], [5]. Task-level success measures the combined capability. Query-level analysis provides a different resolution by following how policy evidence evolves as successive action chunks alter the scene.

B. World models and action-generating video models

World-model policies introduce predicted consequences into action generation, but differ in how prediction is connected to control. RoboDreamer treats a generated video as an intermediate plan and recovers actions through inverse dynamics [8]. Unified World Models instead couples action and future-observation diffusion, while Cosmos Policy jointly produces action chunks, future observations and predicted returns [9], [10]. The first design separates visual planning from action decoding; the latter designs place both outputs in one generative process.

This distinction determines where uncertainty can enter. A staged video policy can accumulate error in the predicted trajectory and in its action decoder, whereas a joint model links action quality to a shared generative representation. Both incur predictive computation that a direct VLA does not require. At the same time, their predicted futures expose intermediate structure that can be checked for consistency. Prior systems establish video-conditioned action generation as a viable policy family. Pairing such a policy with a separately trained direct policy additionally requires deciding when its predictive process is likely to be useful.

C. Uncertainty and runtime failure prediction

That decision depends on uncertainty evidence whose interpretation matches the generating process. Heteroscedastic likelihood learns input-dependent residual variance and represents conditional, or aleatoric, dispersion under the fitted model [14]. Gaussian NLL induces inverse-variance weighting that can progressively underweight poorly fitted regions; Beta-NLL modifies this weighting to reduce the imbalance [15]. Model uncertainty is commonly approximated through disagreement among independently trained predictors [16]. For large robot policies, however, maintaining several models adds training, memory and inference cost. Structured action outputs also make a single global confidence value difficult to interpret: VLA studies report that informative uncertainty varies across action dimensions and over time [7], [6].

A related line of work converts such evidence into temporal failure scores. FAIL-Detect learns a scalar score from successful trajectories and applies time-varying calibrated thresholds [17]. FIPER requires only successful rollouts: it aggregates observation-embedding novelty from RND-OE and grid-based action-chunk entropy over short windows, calibrates both signals and raises an alarm through their logical conjunction [18]. SAFE instead learns failure prediction from temporal VLA features and studies transfer to held-out tasks [19]. Our VLA monitor follows the latter supervised setting but combines physical history, generated chunks, internal flow uncertainty and candidate disagreement in one learned risk score.

Calibration also determines what can be claimed about a detector. Conformal prediction provides finite-sample marginal coverage under exchangeability [20]; FIPER's bound concerns alarms on successful in-distribution rollouts, not failure recall or coverage under arbitrary shift. Moreover, prior VLA detectors monitor one policy distribution. They do not establish that features or thresholds calibrated for a direct action model transfer to a predictive policy with different outputs and error mechanisms.

III. Problem Formulation

We consider a language-conditioned manipulation episode $e \sim \mathcal{Q}$, where the evaluation distribution $\mathcal{Q}$ may differ from the training distribution through object appearance, scene configuration or task instruction. At control time t, the robot has visual observations $o_{0:t}$, proprioceptive states $q_{0:t}$, previously executed actions $a_{0:t-1}$ and an instruction ℓ. The evidence available to a causal controller is

$$E_t = (o_{0:t}, q_{0:t}, a_{0:t-1}, \ell) \,. \tag{1}$$

The episode terminates at task completion or when its action budget is exhausted. Its outcome is $Y_e \in \{0,1\}$, with $Y_e = 1$ denoting successful completion.

The system contains two frozen action-generating policies indexed by $b \in \{\mathrm{V}, \mathrm{W}\}$. The VLA policy V maps the current evidence directly to an action chunk. The video-world-model policy W processes the evidence with a video-prediction backbone and decodes an action chunk from its intermediate predictive context. Querying branch b returns

$$(A_t^b, u_t^b) = \pi_b(E_t), \qquad A_t^b \in \mathbb{R}^{H_b \times d_a}, \quad u_t^b \in \mathbb{R}^{d_b}, \tag{2}$$

where H_b is the proposed horizon, d_a is the action dimension and u_t^b contains the uncertainty evidence exposed by that branch. The two uncertainty vectors need not share a dimension, scale or probabilistic interpretation.

At each decision point, the controller queries V and may additionally query W. Let $n_t^b \in \{0,1\}$ indicate whether branch b is queried, with $n_t^{\mathrm{V}} = 1$. From the proposals available at time t, a routing rule ρ selects a branch b_t and an executed prefix length $m_t \in \{1, \ldots, H_{b_t}\}$. The applied controls are $A_t^{b_t}[1 : m_t]$. Both the selected chunk and the evidence used to select it must correspond to the same environment state E_t; once an action changes the state, an unexecuted proposal cannot be treated as a current decision.

Let c_b be the measured cost of querying branch b, with the predictive branch generally satisfying $c_{\mathrm{W}} > c_{\mathrm{V}}$. The planning cost of episode e is

$$C_e(\rho) = \sum_t \sum_{b \in \{\mathrm{V},\mathrm{W}\}} c_b n_t^b. \tag{3}$$

The desired router maximizes task success on the evaluation distribution while respecting an expected planning budget B:

$$\max_\rho \ \mathbb{E}_{e\sim\mathcal{Q}}[Y_e] \quad \text{subject to} \quad \mathbb{E}_{e\sim\mathcal{Q}}[C_e(\rho)] \le B. \tag{4}$$

Problem. Given the frozen policies π_{V} and π_{W}, their branch-native uncertainty evidence and calibration trajectories, construct a causal routing rule ρ that selects executable action prefixes at query time and improves out-of-distribution task success without exceeding the planning budget.

IV. Methodology

A. System overview

U-VOWEL connects two independently trained action-generating policies through the interface in Eq. (2). Both branches receive the same language instruction from a synchronized decision state, but retain their native observations, action representations and generative processes. The direct VLA forms the default path. The video-world-model policy is queried only when the VLA monitor raises an alarm. Figure 2 follows one such decision from observation to execution.

Each policy first produces an action proposal and uncertainty measurements tied to its own prediction process. A branch-specific temporal model converts those measurements into a failure score. The controller then accepts one proposal, shortens VLA execution or temporarily suppresses repeated calls to the predictive branch. This separation prevents predicted variance and sample disagreement from being treated as calibrated probabilities of task failure.

B. Direct VLA policy

1) Action generation: The direct branch modifies SimVLA [4]. A frozen SmolVLM-500M encoder processes the language instruction and current agent and wrist RGB views. A flow-matching action transformer combines this representation with an eight-dimensional proprioceptive state and generates a normalized chunk of $H = 10$ seven-dimensional Cartesian actions. For an expert chunk $A^\star$, Gaussian noise ϵ and flow time $\gamma = 0.001 + 0.999\tilde{\gamma}$, with $\tilde{\gamma} \sim \mathrm{Beta}(1.5, 1)$, training constructs

$$X_\gamma = (1-\gamma)A^\star + \gamma\epsilon, \qquad V_\gamma^\star = \epsilon - A^\star. \tag{5}$$

The action transformer predicts the velocity field from (X_γ, γ), then inference integrates that field from noise to an action chunk. The vision-language encoder remains frozen during uncertainty fine-tuning; the action transformer and its output heads are trainable.

2) Heteroscedastic velocity uncertainty: We replace the original velocity output with parallel velocity and variance heads. For action index i, the second head emits $r_{\theta,i}$, which is mapped to a positive conditional variance,

$$\sigma_{\theta,i}^2 = \mathrm{softplus}(r_{\theta,i}) + \varepsilon_{\mathrm{n}}, \qquad \varepsilon_{\mathrm{n}} = 10^{-6}. \tag{6}$$

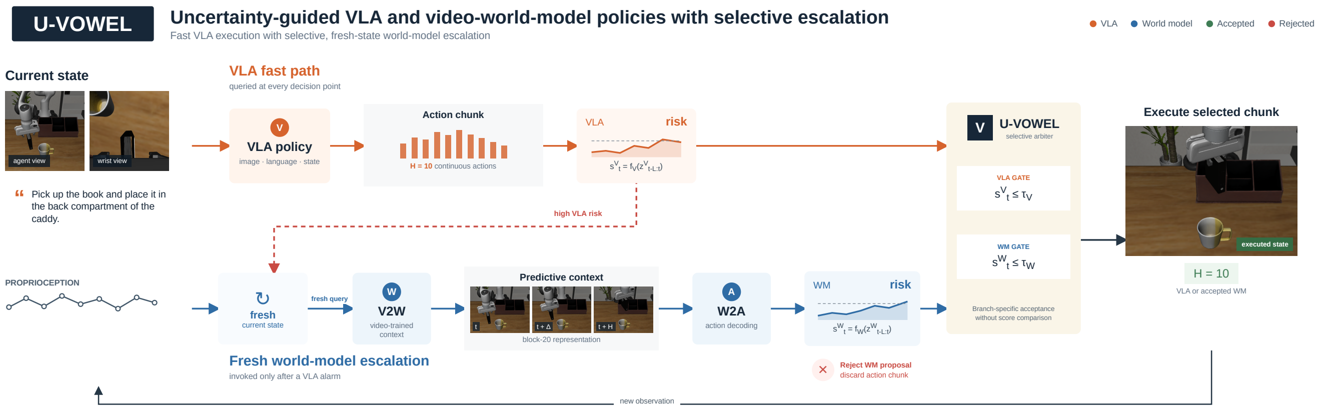

Fig. 2. U-VOWEL execution loop. The VLA supplies the default action chunk and a branch-specific failure score. A VLA alarm makes the predictive branch compute a fresh video-conditioned context and decode an alternative action chunk from the current state. The visual frames represent the V2W training target; the deployed branch passes an intermediate representation to W2A without decoding a pixel-space rollout. The two scores are evaluated against separate thresholds and are never compared numerically.

The modified policy is fine-tuned with an element-wise Beta-NLL objective [15]:

$$\mathcal{L}_{\mathrm{V}} = \frac{1}{|\mathcal{I}|} \sum_{i\in\mathcal{I}} \mathrm{sg}\Big[\big(\sigma^2_{\theta,i}\big)^{\beta}\Big] \times \left[\frac{(V_{\theta,i} - V^{\star}_{\gamma,i})^2}{2\sigma^2_{\theta,i}} + \frac{1}{2}\log\sigma^2_{\theta,i} + \kappa\right], \qquad \beta = 0.5. \tag{7}$$

Here sg stops gradients through the Beta-NLL weight, $\mathcal{I}$ spans action timesteps and dimensions and $\kappa = -\frac{1}{2}\log\varepsilon_{\mathrm{n}}$. This constant keeps the bracketed term nonnegative for $\sigma^2_{\theta,i} \geq \varepsilon_{\mathrm{n}}$. The residual term weights velocity error by predicted precision, while the logarithmic term penalizes indiscriminate variance inflation. The detached Beta-NLL factor reduces the attenuation of high-variance examples without adding another gradient path through the variance weight.

At inference, variance is retained along the Euler trajectory. A diagonal first-order approximation accumulates

$$P_{\gamma+\Delta\gamma} = P_{\gamma} + (\Delta\gamma)^2\sigma^2_{\theta}(X_{\gamma}, \gamma). \tag{8}$$

Each query draws nine independently seeded $H = 10$ chunks from one encoded observation. Candidate zero is the main proposal; eight alternatives provide uncertainty evidence. A 49-dimensional record combines the main candidate's last-step and path-integrated variance, denoising dynamics, chunk geometry and dispersion across all nine samples. Separately, a seven-dimensional candidate- disagreement descriptor is computed from the eight normalized alternatives: centered and pairwise chunk distances, overall dispersion and translation, rotation, gripper and flattened-coordinate dispersion. This descriptor is not FIPER's entropy-based ACE. The deployed monitor retains eight of the 49 dimensions through a ranking performed on seen train/validation data, covering initial denoising magnitude, update-direction flips, path variance and action- sample dispersion. None of these inputs contains a task identifier, reward or future observation.

C. Predictive video–action policy

1) V2W–W2A generation: The predictive branch separates a video-trained representation from action decoding. Its video-to-world component (V2W) adapts a Cosmos Predict2 backbone through rank-256 LoRA and trainable adaptive-normalization parameters [21], [10]. V2W is trained to predict future video from five 480×640 agent-view frames sampled from recent observation history and the language embedding. The deployed policy does not complete or decode that visual rollout. It stops at the first V2W denoising evaluation and extracts a hidden representation after transformer block 20 as the predictive context C_t. The world-to-action component (W2A) is a separate diffusion transformer trained while V2W remains frozen. It conditions on C_t, the ten-dimensional robot state and the video noise level, then generates 60 ten-dimensional tokens. Translation, six-dimensional rotation and gripper outputs are decoded to the seven-dimensional environment action.

W2A uses the interpolation and velocity target in Eq. (5), with $\gamma = 0.001 + 0.999\tilde{\gamma}$ and $\tilde{\gamma} \sim \mathrm{Beta}(1, 1)$, and minimizes

$$\mathcal{L}_{\mathrm{W2A}} = 10\,\mathbb{E}_{\gamma,\epsilon}\left[\Big\|V_{\omega}(X_{\gamma}, \gamma; q_t, C_t, \sigma) - V^{\star}_{\gamma}\Big\|_2^2\right], \tag{9}$$

where σ is the V2W noise level. During training, the observation condition is dropped with probability 0.2. This follows the flow-matching formulation [22]. Although W2A predicts a long sequence, the controller executes at most ten actions before observing the environment again.

Repeated W2A samples expose action uncertainty without repeating the expensive video pass. In the K1 configuration, one V2W context produces eight W2A candidates. Their per-step and per-dimension variance, pairwise mean-squared distance, receding-plan disagreement and denoising dynamics form the K1 uncertainty vector. Candidate zero remains the executable proposal; the other samples provide evidence and do not compete for control. K3 instead generates three V2W contexts. The main context produces eight W2A candidates and each additional context produces one fixed-seed candidate, giving ten action chunks for measuring sensitivity to the sampled V2W representation. Thus K1 and K3 name the number of V2W contexts, not the number of executed chunks or the monitor history.

2) Residual uncertainty in V2W: Separately, we train a lightweight residual-variance head on the frozen V2W backbone. Let $h_{20,i}$ be a future token at block 20 and let ρ_i^2 be the mean squared clean-latent residual over its 16 channels. A calibration table supplies a positive base variance $b_{q(\sigma)}$ for the current video-noise bin. The head predicts a token-wise correction from $h_{20,i}$ and an embedding of $\log \sigma$:

$$\begin{aligned} \ell_i &= \mathrm{clip}\big(\log b_{q(\sigma)} + \Delta_\phi(h_{20,i}, \log \sigma), -8, 5\big)\,, \\ v_i &= \exp(\ell_i). \end{aligned} \tag{10}$$

Only the variance head and noise embedding are optimized. With $w_i = 1$ for future tokens and zero for observed conditioning tokens, the loss is

$$\mathcal{L}_{\text{V2W-NLL}} = \frac{1}{2} \frac{\sum_i w_i \left(\rho_i^2 / v_i + \ell_i\right)}{\sum_i w_i + \varepsilon_{\text{n}}}. \tag{11}$$

This head estimates conditional latent-residual scale. It does not by itself measure task failure or complete epistemic uncertainty. Its summaries enter the uncertainty-feature ablations. The selected K1 controller uses the action-uncertainty profile described above and therefore does not depend on V2W variance-head features.

D. Branch-specific temporal risk

FIPER [18] provides the methodological starting point for runtime monitoring: RND-OE measures novelty in the policy observation embedding, while grid-based action-chunk entropy (ACE) measures dispersion among repeated action samples. The two scores are accumulated over short windows, calibrated from successful rollouts and combined by a logical AND. Our monitors instead learn a single branch-specific score from seen successful and failed rollouts, using the generated action evidence and its recent history. The VLA candidate- disagreement descriptor above is a seven-value feature vector, not FIPER ACE; the VLA monitor also uses neither RND nor direct observation embeddings.

Let R_t^b denote the normalized query record built for branch b. The record may contain sequential action or uncertainty tokens, a bounded history and current summary features. Each monitor computes

$$s_t^b = \mathrm{sigmoid}\big(f_b(R_t^b)\big)\,, \qquad b \in \{\mathrm{V}, \mathrm{W}\}. \tag{12}$$

The records and functions are branch-specific because the two generative models expose different evidence.

The VLA monitor uses a late-fusion Transformer. Inputs are standardized using statistics fitted only on the training split. Its sequence path contains 16 history tokens of dimension 21—proprioception (8), the previous executed action (7) and six candidate-disagreement values—followed by ten 7D tokens from the normalized main proposal. Separate linear projections map both token types to width 128. A learned CLS token and positional embeddings precede a three-layer, four-head Transformer with 512-dimensional feed-forward blocks. Startup histories are zero-padded and processed without an attention mask.

The static 51D path concatenates 28 main-chunk statistics (first, mean, standard deviation and endpoint change), the seven candidate-disagreement values, current 8D proprioception and the selected eight uncertainty features. A linear–GELU projection maps it to width 128. Its output is concatenated with the Transformer CLS output, normalized and passed through a 128D hidden layer to the risk logit. Neither task identity nor timestep is an input.

The selected K1 world-model monitor uses an eight-query history and a two-layer GRU. At each query it receives 74 scalar W2A uncertainty features, including candidate variance, pairwise disagreement, plan overlap and denoising summaries. Sixteen temporal action-uncertainty tokens preserve the per-step structure of those statistics. The GRU and static branch have width 128 and their fused representation passes through a 64-dimensional latent layer to the risk logit. V2W residuals, multi-context disagreement and candidate centrality define richer profiles used in the ablations, but are absent from this K1 feature contract.

The monitors are trained as binary classifiers from rollout outcomes. A query in a failed rollout receives a failure target even when that query did not cause the eventual failure. For the VLA monitor, let $F_e = 1 - Y_e$ denote episode failure, and let N_0 and N_1 be the numbers of successful and failed training rows. Its positive-class weight is $p = \max(1, N_0 / \max(1, N_1))$, and the implemented objective is

$$\mathcal{L}_{\text{risk}}^{\text{V}} = -\frac{1}{N} \sum_{e,t} \left[p F_e \log s_{e,t}^{\text{V}} + (1 - F_e) \log(1 - s_{e,t}^{\text{V}}) \right]. \tag{13}$$

Consequently, s_t^b is an empirical warning score for the observed episode prefix, not a causal probability that executing A_t^b will fail.

Each score is compared only with its branch threshold. VLA operating points are computed on a held-out seen validation split: one maximizes row-level F1 using both classes, while success-only alternatives use empirical quantiles of successful row scores. The operating point is selected according to the desired seen false-alarm/detection trade-off and frozen before unseen evaluation. An episode alarms if any query crosses it. This is empirical row-score calibration, not an episode-level conformal guarantee.

The world-model threshold is calibrated from the episode maximum score on successful held-out rollouts. For n calibration episodes with maxima $g_e = \max_t s_{e,t}^{\text{W}}$, its level-$\alpha$ threshold is the corrected empirical quantile

$$\tau_{\text{W}}(\alpha) = g_{(\min\{n, \lceil (n+1)(1-\alpha) \rceil\})}, \tag{14}$$

where $g_{(k)}$ is the k-th order statistic. This controls the marginal episode-level false-alarm rate under the usual exchangeability assumption [20]; it does not guarantee calibration after arbitrary sequential distribution shift.

Algorithm 1 U-VOWEL selective execution for one episode

Require: policies $\pi_{\mathrm{V}}, \pi_{\mathrm{W}}$, monitors $f_{\mathrm{V}}, f_{\mathrm{W}}$, thresholds $\tau_{\mathrm{V}}, \tau_{\mathrm{W}}$, horizons $H > h$, latch span C

$c \leftarrow 0$ ▷ remaining ineligible environment actions
while episode active and action budget remains **do**
  acquire current evidence E_t
  $(A^{\mathrm{V}}, u^{\mathrm{V}}) \leftarrow \pi_{\mathrm{V}}(E_t)$; $s^{\mathrm{V}} \leftarrow f_{\mathrm{V}}(R_t^{\mathrm{V}})$
  if $s^{\mathrm{V}} \leq \tau_{\mathrm{V}}$ **then**
    $m \leftarrow H$ if $c = 0$, else $\min(H, c)$
    $n \leftarrow$ EXECUTEPREFIX(A^{V}, m); $c \leftarrow \max(c - n, 0)$
  else if $c > 0$ **then**
    $n \leftarrow$ EXECUTEPREFIX$(A^{\mathrm{V}}, \min(h, c))$; $c \leftarrow \max(c - n, 0)$
  else
    $(A^{\mathrm{W}}, u^{\mathrm{W}}) \leftarrow \pi_{\mathrm{W}}(E_t)$ ▷ fresh query from the current state
    $s^{\mathrm{W}} \leftarrow f_{\mathrm{W}}(R_t^{\mathrm{W}})$
    if $s^{\mathrm{W}} \leq \tau_{\mathrm{W}}$ **then**
      EXECUTEPREFIX(A^{W}, H)
    else
      DISCARDPROPOSAL(A^{W})
      $c \leftarrow C$
      $n \leftarrow$ EXECUTEPREFIX$(A^{\mathrm{V}}, \min(h, c))$; $c \leftarrow \max(c - n, 0)$
    end if
  end if
end while

E. Selective execution

Algorithm 1 combines the two monitors without comparing their score magnitudes. The state variable c is the number of environment actions remaining before the world-model branch becomes eligible after a rejected proposal. When $c = 0$, a VLA alarm triggers a fresh V2W–W2A query from the same evidence E_t. An accepted world-model proposal supplies the next action prefix. If its own monitor also alarms, the proposal is discarded, the controller executes a shorter prefix of the already available VLA chunk and starts the latch. The selected K1 controller uses $H = 10$, $h = 5$ and $C = 50$.

While the latch is active, the VLA is still queried and scored at every decision point. A high VLA score no longer invokes V2W–W2A and receives the short horizon h. A low score may use the ordinary horizon H, capped at latch expiry so that eligibility is reconsidered at the next state. The latch is decremented by actions actually applied, including early termination within a chunk. Once it expires, the next VLA alarm runs new world-model inference. The rejected action buffer is never reused; its score remains only in the monitor history and audit log.

V. EXPERIMENTAL PROTOCOL

A. Evaluation suites

- **LIBERO in-distribution evaluation:** standard LIBERO suites will measure task competence before perturbation. Unperturbed LIBERO-Goal episodes will also provide the in-distribution reference for the corresponding LIBERO-PRO tasks.
- **LIBERO-PRO evaluation:** the official Goal-Language, Goal-Object, Goal-Swap and Goal-Task suites contain ten tasks with 50 initial states per task. The same 2,000 episode identities will be used for every policy and controller configuration.
- **IsaacLab evaluation:** the Franka environment will separate in-distribution trials from controlled changes in object identity, category, geometry, material, pose, lighting, background, camera and combinations of these factors. Asset-family-disjoint splits will support the unknown-object analysis.
- **Physical evaluation:** frozen simulation-selected checkpoints, monitors and thresholds will be transferred to the real Franka setup. Trial blocks will cover familiar objects, novel identities within known categories and held-out object categories.

B. Baselines and comparison policy

- Standalone baselines comprise the base VLA, uncertainty-instrumented VLA, base video-world-model policy and uncertainty-instrumented K1 and K3 video branches.
- Routing baselines comprise always-VLA, always-world-model, random budget-matched selection, static task-level selection, raw-uncertainty thresholding, a single-gate controller and the complete U-VOWEL controller.
- Reproduced state-of-the-art policy baselines will use the same observations, prompts, action interface and episode manifests whenever their implementations permit. Published results obtained under different protocols will be reported in a separate comparison block and excluded from matched significance tests.
- The IsaacLab environment is project-specific. Its comparison will use strong policy families adapted to the same data and control interface rather than presenting an unmatched public-leaderboard claim.

C. Metrics and statistical analysis

- Primary task metrics are episode success, per-task success and suite-level success. Paired outcome tables will count shared successes, shared failures and one-policy-only successes on identical episodes.
- Offline monitor metrics include AUROC, AUPRC, risk–coverage behavior, failure detection, success false alarms, detection by query number and warning lead time where the labels support it.
- Controller metrics include branch-query counts, escalation rate, world-model acceptance, mutual rejection, useful and harmful outcome reversals and latch activity.
- Efficiency metrics include branch inference time, monitor overhead, all-episode wall time, successful-episode wall time, throughput and peak GPU memory. Success-conditioned time will always be reported beside the all-episode statistic.
- Confidence intervals and paired tests will respect the repeated task–initial-state structure. Results will be separated by seed before any aggregate is formed.

D. Sim-to-real protocol and safety

- Simulation and hardware interfaces will be checked for camera geometry, color conversion, state representation, control frequency, workspace frame and action scaling.

- Policy order will be randomized within object and condition blocks. Checkpoints, prompts, thresholds and stopping rules will remain frozen during the reported evaluation.
- Outcomes will distinguish success, drop, collision, timeout, automatic stop and human intervention. Independent workspace, velocity, force, watchdog and emergency-stop limits remain outside the learned controller.

VI. Results

A. Ablation studies

a) Uncertainty evidence.: The offline study will compare each uncertainty family alone and in combination. For the VLA, this includes predicted variance, candidate disagreement, action statistics and denoising-path summaries. For the video branch, it includes residual variance, V2W summaries, W2A disagreement, overlap, churn and denoising diagnostics. Tables will report held-out ranking and alarm metrics, while feature-group heatmaps will show which signals remain useful across tasks.

b) Temporal history.: Memoryless scoring will be compared with several history lengths and recurrent monitor configurations. K1 and K3 will be evaluated as predictive-context variants rather than treated as equivalent-cost models. Detection curves by query number will show whether additional history improves early warnings or only identifies failures late in the episode.

c) Thresholds.: Branch-specific alpha sweeps will report detection against successful-episode false alarms. The analysis will compare global branch thresholds with any task-conditioned operating point, show calibration and held-out behavior separately and identify the operating points used online. Threshold sensitivity will also be measured through downstream task success and world-model query rate.

d) Risk models and evidence fusion.: Memoryless and temporal risk models will be compared under the same feature contract and data split. Single-source, scalar-fusion and richer feature-fusion variants will separate the contribution of the model architecture from the contribution of its inputs. Offline results will include AUROC, AUPRC, risk–coverage curves, detection, false alarms and warning time.

e) Online policy fusion.: The online ablation will compare standalone policies, random and static routing, raw-uncertainty routing, VLA-gate-only, world-model-gate-only and dual-gate controllers. It will also vary execution prefix, latch length, K1 versus K3 and fresh-state versus cached proposals as a diagnostic. Success, paired outcome reversals, branch use and latency will determine whether an offline monitor improvement produces a useful control decision.

B. Simulation results

a) LIBERO in-distribution results.: A main table will compare U-VOWEL, its standalone branches and reproducible state-of-the-art baselines on standard LIBERO. Results will include overall, suite and task success, paired complementarity and latency. Published numbers from unmatched protocols will occupy a separate block with their evaluation differences stated explicitly.

b) LIBERO-PRO in-distribution reference.: Unperturbed LIBERO-Goal episodes will establish the reference performance for the ten tasks before applying LIBERO-PRO perturbations. This comparison will separate a weak base policy from degradation caused by the evaluation shift and will report branch-specific failure patterns on matched episode identities.

c) LIBERO-PRO out-of-distribution results.: Goal-Language, Goal-Object, Goal-Swap and Goal-Task will be reported overall, by suite and by task. Success heatmaps, paired win–loss matrices and controller event plots will identify where the branches are complementary, where routing causes harmful switches and where both policies lack the required capability.

d) IsaacLab in-distribution and OOD results.: The simulation study will first validate each branch on familiar assets and poses, then evaluate controlled object and scene shifts. Tables will compare the world model, VLA, U-VOWEL and adapted strong baselines using success, final pose error, drop or collision rate, alarm quality, branch use and latency. Results will be grouped by held-out object family to support the unknown-object claim.

e) Success–compute trade-off.: A common plot will place standalone policies and routing variants on a success–compute frontier using measured wall time and world-model invocation rate. All-episode and successful-episode timing will be shown separately, with per-query timing used to explain the observed episode-level cost.

C. Sim-to-real results

a) World-model policy.: The first hardware result will report the standalone video-world-model branch on familiar and held-out objects. The analysis will include success, prediction and action-decoding failures, stopping events, latency and the change from its matched IsaacLab performance.

b) VLA policy.: The second result will report the standalone VLA under the same randomized trial blocks. Per-object success, grasp or placement failures, uncertainty alarms, episode time and the simulation-to-real gap will be compared with the world model without changing the trial manifest.

c) Fused U-VOWEL controller.: The final hardware comparison will evaluate the frozen fused controller. Tables will report task success, useful and harmful escalations, world-model invocation and acceptance, mutual rejection, automatic stops and latency. Qualitative sequences will illustrate successful escalation, unnecessary escalation and cases where neither branch provides a valid action.

d) Cross-domain summary.: The closing result will compare the ranking of the three systems across IsaacLab and the physical robot. The analysis will determine whether monitor behavior, policy complementarity and the selected success–compute operating point transfer beyond simulation.

VII. Discussion

A. Policy complementarity under distribution shift

- The discussion will relate gains to matched episodes solved by only one branch, distinguishing genuine complementarity from changes in execution horizon or query frequency.
- Suite- and task-level patterns will be used to explain which semantic, object and control shifts favor direct action generation or predictive visual reasoning.
- Cases where routing fails will be separated into late alarms, false alarms, rejected useful proposals, prediction errors, action-decoder errors and tasks outside the capability of both branches.

B. Uncertainty as a control signal

- Offline ranking quality will be interpreted through the online decisions it enables. A higher AUROC is relevant only when the selected threshold produces timely interventions at an acceptable false-alarm and compute cost.
- The usefulness of individual uncertainty families will be discussed without assigning a stronger epistemic interpretation than their training objective supports.
- Threshold sensitivity and transfer across task, simulator and hardware conditions will define the practical calibration limits of the current system.

C. Efficiency and deployment implications

- The success–compute frontier will show when selective predictive inference preserves a useful portion of VLA speed and when repeated escalation removes that advantage.
- Query-level and successful-episode timing will be used together to avoid attributing latency differences to policies that simply terminate earlier or fail more often.
- The controller will be positioned as a way to operate complementary learned policies under a budget, not as a replacement for missing policy capability or independent robot safety mechanisms.

D. Limitations

- Episode-level monitor labels do not identify the exact action that makes a trajectory unrecoverable and can reward alarms that occur after the decisive failure.
- Heteroscedastic variance heads model conditional residual dispersion and do not provide a complete estimate of model uncertainty.
- Thresholds are distribution dependent. Their reliability under new tasks, embodiments and hardware conditions must be measured rather than assumed.
- Published state-of-the-art results are not directly comparable when observations, checkpoints, action horizons or episode manifests differ. Matched local reruns remain the primary evidence.
- IsaacLab and real-robot results are limited by available object families, trial counts and safety constraints. Routing cannot recover a skill absent from both policies.

VIII. Conclusion

The conclusion will return to the paper's central question: whether uncertainty can make a fast VLA and a predictive video-world-model policy more effective as a selective system than as standalone policies. It will summarize the final evidence in three parts: the quality and timing of branch-specific failure signals, the success–compute behavior of online fusion and the transfer of policy complementarity from LIBERO and IsaacLab to the physical robot. Exact claims will be limited to completed matched evaluations.

The final paragraph will identify the remaining research directions suggested by the experiments: stronger supervision for warning time, uncertainty that better captures model ignorance, calibration across changing tasks and embodiments and recovery policies that add capabilities beyond those already present in either branch.

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